

LAB 11

Ex 01:

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
int main() {
    string name;
    int age;
    cout << "Enter name: ";
    getline(cin, name);
    cout << "Enter Age: ";
    cin >> age;
    ofstream file;
    file.open("user.txt");
    if (!file) {
        cout << "Could not open file!" << endl;
        return 1;
    }
    file << "Name: " << name << ", Age: " << age << endl;
    file.close();
    cout << "Data saved successfully!";
    return 0;
}
```

Output:

```
✓ TERMINAL
cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ ex1.cpp -o ex1 && "/home/rumaisa-abbasi/Desktop/Lab 11/"ex1
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$ cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ ex1.cpp -o ex1 && "/home/rumaisa-abbasi/Desktop/Lab 11/"ex1
Enter name: Rumaisa
Enter Age: 18
○ Data saved successfully!rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$
```

Task 01:

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
int main() {
    string name;
    int age;
    cout << "Enter name: ";
    getline(cin, name);
    cout << "Enter Age: ";
    cin >> age;
    ofstream file;
    file.open("user.txt", ios::app);
    if (!file) {
        cout << "Could not open file!" << endl;
        return 1;
    }
    file << "Name: " << name << ", Age: " << age << endl;
    file.close();
    cout << "Data saved successfully!";
    return 0;
}
```

Output:

```
Name: Rumaisha, Age: 18
Name: Rumaisha , Age: 18
```

```
✓ TERMINAL
cd "/home/rumaisha-abbasi/Desktop/Lab 11/" && g++ ex1.cpp -o ex1 && "/home/rumaisha-abbasi/De
b 11/"ex1
● rumaisha-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$ cd "/home/rumaisha-abbasi/De
1/" && g++ ex1.cpp -o ex1 && "/home/rumaisha-abbasi/Desktop/Lab 11/"ex1
Enter name: Rumaisha
Enter Age: 18
○ Data saved successfully! rumaisha-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$
```

EX 02:

```
#include <iostream>
#include <fstream>
using namespace std;
int main() {
    ofstream file;
    file.open("Ex2.txt");
    if (!file) {
        cout << "Could not open file for writing!" << endl;
        return 1;
    }
    for (int i = 0; i < 10; i++) {
        file << i << " ";
    }
    file.close();
    ifstream in;
    in.open("Ex2.txt");
    if (!in) {
        cout << "Could not open file for reading!" << endl;
        return 1;
    }
    int num;
    while (in >> num) {
        if (num % 2 == 0) {
            cout << " " << num;
        }
    }
    in.close();
    return 0;
}
```

Output:

```
cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ ex2.cpp -o ex2 && "/home/rumaisa-abbasi/Desktop/Lab 11/"ex2
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$ cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ ex2.cpp -o ex2 && "/home/rumaisa-abbasi/Desktop/Lab 11/"ex2
○ 0 2 4 6 8rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$
```

Task 02:

```
task2.cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
int main() {
    ifstream file("Ex2.txt");
    string line;
    int lineCount = 0;
    if (file.is_open()) {
        while (getline(file, line)) {
            lineCount++;
        }
        file.close();
        cout << "Total number of lines: " << lineCount << endl;
    } else {
        cout << "Error: Could not open file." << endl;
    }
    return 0;
}
```

Output:

```
cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ task2.cpp -o task2 && "/home/rumaisa-abbasi/Desktop/Lab 11/task2
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$ cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ task2.cpp -o task2 && "/home/rumaisa-abbasi/Desktop/Lab 11/task2
Total number of lines: 1
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$
```

Ex 03:

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
class Student {
private:
    int id;
    string name;
    float gpa;
public:
    void input() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        getline(cin, name);
        cout << "Enter GPA: ";
        cin >> gpa;
        cout << "-----" << endl; }
    void writeToFile() {
        ofstream file("student.txt", ios::app);
        if (file.is_open()) {
            file << id << " " << name << " " << gpa << endl;
            file.close();} }
};
int main() {
    Student s;
    for (int i = 1; i <= 5; i++) {
        cout << "Recording Student #" << i << ":" << endl;
        s.input();
        s.writeToFile();
    }
    cout << "All 5 student records have been saved to student.txt!" << endl;
    return 0;
}
```

Output:

```
Recording Student #1:
Enter ID: 101
Enter Name: Rumaisa
Enter GPA: 3.75
-----
Recording Student #2:
Enter ID: 102
Enter Name: Ali
Enter GPA: 3.5
-----
Recording Student #3:
Enter ID: 103
Enter Name: Sara
Enter GPA: 3.2
-----
Recording Student #4:
Enter ID: 104
Enter Name: Zain
Enter GPA: 3.7
-----
Recording Student #5:
Enter ID: 105
Enter Name: Fahad
Enter GPA: 3.0
-----
All 5 student records have been saved to student.txt!
```


Task 03:

```
#include <string>
using namespace std;
class Student {
private:
    int id;
    string name;
    float gpa;
public:
    void input() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        getline(cin, name);
        cout << "Enter GPA: ";
        cin >> gpa;
        cout << "-----" << endl; }
    void writeToFile() {
        ofstream file("student.txt", ios::app);
        if (file.is_open()) {
            file << id << " " << name << " " << gpa << endl;
            file.close();
        } }
    void readFromFile() {
        ifstream file("student.txt");
        if (!file) {
            cout << "No records found!" << endl;
            return; }
        cout << "\n--- All Student Records ---\n";
        while (file >> id >> name >> gpa) {
            cout << "ID: " << id << " | Name: " << name << " | GPA: " << gpa << endl;
        }
        file.close(); }
};

int main() {
    Student s;
    for (int i = 1; i <= 5; i++) {
        cout << "Recording Student #" << i << ":" << endl;
        s.input();
        s.writeToFile(); }
    cout << "All 5 student records have been saved to student.txt!" << endl;
    s.readFromFile();
    return 0;
}
```

Output:

```
1/ ~ && g++ tasks.cpp -o tasks && ./tasks && /home/...
-----
Recording Student #2:
Enter ID: 102
Enter Name: ali
Enter GPA: 3.5
-----
Recording Student #3:
Enter ID: 103
Enter Name: sara
Enter GPA: 3.5
-----
Recording Student #4:
Enter ID: 104
Enter Name: zain
Enter GPA: 3.4
-----
Recording Student #5:
Enter ID: 105
Enter Name: fahad
Enter GPA: 3.2
-----

--- All Student Records ---
ID: 101 | GPA: 3.7 | Name: rumaisa
ID: 102 | GPA: 3.5 | Name: ali
ID: 103 | GPA: 3.5 | Name: sara
ID: 104 | GPA: 3.4 | Name: zain
ID: 105 | GPA: 3.2 | Name: fahad
```


EX 04:

```
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;
class Student {
public:
    int id;
    char name[50];
    void setData(int i, const char* n) {
        id = i;
        strncpy(name, n, 49);
        name[49] = '\0';
    }
};
int main() {
    Student s[3];
    s[0].setData(101, "Ali");
    s[1].setData(102, "Sara");
    s[2].setData(103, "Zayn");
    ofstream file("student.dat", ios::binary);
    if (!file) {
        cout << "Error opening file!" << endl;
        return 1;
    }
    for (int i = 0; i < 3; i++) {
        file.write((char*)&s[i], sizeof(s[i]));
    }
    file.close();
    cout << "Multiple binary records written successfully." << endl;
    return 0;
}
```

Output:

```
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$ cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ ex4.cpp -o ex4 && "/home/rumaisa-abbasi/Desktop/Lab 11/"ex4
Multiple binary records written successfully.
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$
```

Task 04:

```
#include <iostream>
#include <fstream>
using namespace std;
class Student {
public:
    int id;
    char name[50];
};
int main() {
    Student s;
    ifstream file("student.dat", ios::binary);
    if (!file) {
        cout << "Error: Could not open file!" << endl;
        return 1;
    }
    cout << "--- Student Records from Binary File ---" << endl;
    while (file.read((char*)&s, sizeof(s))) {
        cout << "ID: " << s.id << " | Name: " << s.name << endl;
    }
    file.close();
    return 0;
}
```

Output:

```
p/Lab 11/ task4
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$ cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ task4.cpp -o task4 && "/home/rumaisa-abbasi/Desktop/Lab 11/task4"
--- Student Records from Binary File ---
ID: 101 | Name: Ali
ID: 102 | Name: Sara
ID: 103 | Name: Zayn
rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$
```

Ex 05:

```
ex5.cpp
1  #include <iostream>
2  #include <fstream>
3  using namespace std;
4  int main() {
5      try {
6          ifstream file("data.txt");
7          if (!file) {
8              throw "Error: File does not exist!";
9          }
10         cout << "File opened successfully." << endl;
11     }
12     catch (const char* msg) {
13         cout << msg << endl;
14     }
15     return 0;
16 }
```

Output:

```
D 11/ ex5
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$ cd "/home/rumaisa-abbasi/Desktop/Lab 11/" && g++ ex5.cpp -o ex5 && "/home/rumaisa-abbasi/Desktop/Lab 11/"ex5
Error: File does not exist!
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$
```

Task 05:

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
int main() {
    string filename = "students.txt";
    try {
        ofstream outFile(filename);
        if (!outFile) {
            throw "Error: Failed to create file for writing!";
        }
        outFile << 101 << " Ali" << endl;
        outFile << 102 << " Sara" << endl;
        outFile.close();
        ifstream inFile(filename);
        if (!inFile) {
            throw "Error: File not found for reading!";
        }
        int id;
        string name;
        cout << "Reading data from file..." << endl;
        while (inFile >> id >> name) {
            cout << "ID: " << id << ", Name: " << name << endl;
        }
        if (!inFile.eof() && inFile.fail()) {
            throw "Error: Corrupted data format found in file!";
        }
        inFile.close();
    }
    catch (const char* errorMsg) {
        cout << "EXCEPTION CAUGHT: " << errorMsg << endl;
    }
    return 0;
}
```

Output:

```
p7/Lab 11/ task5
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$ cd "/home/
1/" && g++ task5.cpp -o task5 && "/home/rumaisa-abbasi/Desktop/Lab 11
Reading data from file...
ID: 101, Name: Ali
ID: 102, Name: Sara
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab 11$
```

Final Task:

```
code.cpp
#include <iostream>
#include <fstream>
#include <string>
using namespace std;
class Student {
public:
    int id;
    string name;
    void input() {
        cout << "Enter ID: "; cin >> id;
        cin.ignore();
        cout << "Enter Name: "; getline(cin, name); }
    void display() {
        cout << "ID: " << id << " | Name: " << name << endl; }
};

void addStudent() {
    Student s;
    s.input();
    ofstream file("data.txt", ios::app);
    if (!file) throw "Error: Could not open file for writing!";
    file << s.id << " " << s.name << endl;
    file.close();
    cout << "Record saved successfully!\n";
}

void viewStudents() {
    Student s;
    ifstream file("data.txt");
    if (!file) throw "Error: File not found!";
    cout << "\n--- Student List ---\n";
    while (file >> s.id >> s.name) {
        s.display(); }
    file.close();
}

void searchStudent() {
    int searchId;
    cout << "Enter ID to search: "; cin >> searchId;
    ifstream file("data.txt");
    if (!file) throw "Error: File not found!";
    Student s;
    bool found = false;
    while (file >> s.id >> s.name) {
        if (s.id == searchId) {
            cout << "Record Found: ";
            s.display();
            found = true;
            break; }
    }
    if (!found) cout << "Student ID not found.\n";
    file.close();
}

int main() {
    int choice;
    while (true) {
        try {
            cout << "\n1. Add Student\n2. View All\n3. Search\n4. Exit\nChoice: ";
            cin >> choice;
            if (choice == 1) addStudent();
            else if (choice == 2) viewStudents();
            else if (choice == 3) searchStudent();
            else if (choice == 4) break;
            else cout << "Invalid choice!\n";
        }
        catch (const char* msg) {
            cout << "EXCEPTION: " << msg << endl; }
    }
    return 0;
}
```


Output:

```
17  && g++ Findtask.cpp -o Findtask &&
1  Add Student
2  View All
3  Search
4  Exit
Choice: 1
Enter ID: 101
Enter Name: Rumaisa
Record saved successfully!

1. Add Student
2. View All
3. Search
4. Exit
Choice: 2

--- Student List ---
ID: 101 | Name: Rumaisa

1. Add Student
2. View All
3. Search
4. Exit
Choice: 3
Enter ID to search: 101
Record Found: ID: 101 | Name: Rumaisa
```